



LCI Advanced Workshop 2025: Power Usage & Monitoring

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Power Usage Overview

- Introduction and Objectives
 - What is the goal?
 - Who will use these?
- Fundamentals
 - Enabling Power Collection

What is the Goal?

Things Admins Care About:

- Utilization Metrics

Things Leadership Cares About:

- Environmental & sustainability mandates
- Energy Costs

Things Users Care About:

- Users will probably not care

Who Uses Power Monitoring?

Power monitoring might help inform higher-level decision-making centered around *where* to keep equipment.

Being informed of power costs—aggregate and point-in-time energy use—can help with capacity planning for future sites or expansion in an existing site. It can help determine how facilities management might distribute power cabling.

Enabling Power Collection

Ensure the dependencies are present for the compilation of slurm:

```
# dnf install -y libcurl-devel

./configure: WARNING: unable to locate freeipmi installation
(libipmonitoring/libfreeipmi)
checking for freeipmi installation...
# dnf install -y freeipmi-devel
```

Enable your preferred power reporting mechanism.

```
--with-freeipmi=PATH    Specify path to freeipmi installation
```

Note, even if this is absent, an identified freeipmi will be built into the binary, as evidenced by our build.

Still, it might be best for `--with-freeipmi` to be specified, so that the build cannot complete without its presence.

Enabling Power Collection

IPMI is the most likely reporting system, since it will be built into data-center bare metal hosts, already.

Other systems like rapl or XCC are too specific to address here, but they are available.

A full list exists in the docs: https://slurm.schedmd.com/acct_gather.conf.html

Virtual Machines like the ones in this lab will not have access to power data, nor will most in the field. Power collection telemetry for these kinds of deployments might therefore depend on higher-level collection from the VM hosts themselves, but this is not supported by Slurm or otherwise, so your implementation will vary.

Enabling Power Collection

`acct_gather_energy`

Power collection requires edits to both `slurm.conf` as well as a separate `acct_gather.conf`.

In `slurm.conf`:

```
AcctGatherEnergyType=acct_gather_energy/ipmi
JobAcctGatherType=jobacct_gather/cgroup
AcctGatherNodeFreq=60
```

More plugins exist than will likely exist on your system, so you do not need to include all the possible options among:

- `AcctGatherEnergyType` (plugin type=`acct_gather_energy`)
- `AcctGatherInterconnectType` (plugin type=`acct_gather_interconnect`)
- `AcctGatherFilesystemType` (plugin type=`acct_gather_filesystem`)
- `AcctGatherProfileType` (plugin type=`acct_gather_profile`)

This information gets saved alongside your other accounting info in your `mysqldb`:
`slurmd -> slurmctld -> mysqldb`

Enabling Power Collection

With just this in `acct_gather.conf`, we should already be able to get the following information from power usage:

```
NodeName=sg035 Arch=x86_64 CoresPerSocket=24  
  CPULloc=26 CPUEfctv=48 CPUTot=48 CPULoad=3.61  
...  
  CurrentWatts=1160 AveWatts=692
```

Fun fact: *Point-in-time usage is fully accessible through Slurmrestd API calls.*

The amount of sensors are not just these two—but these are what show up in `scontrol show node`. A much more comprehensive collection is available according to your IPMI's capabilities.

Enabling Power Collection

On the job-level, completed jobs will report the amount of energy they used:

JobId=14939 ArrayJobId=33004 ArrayTaskId=16367 JobName=autodockvina-run

...

RunTime=00:01:10 TimeLimit=04:00:00 TimeMin=N/A

ReqTRES=cpu=4,mem=2G,node=1,billing=4

AllocTRES=cpu=4,mem=2G,energy=29344,node=1,billing=4

The Energy is reported in J (Joules).

Enabling Power Collection

In `acct_gather.conf`:

```
EnergyIPMIFrequency=10           #the number of seconds between samples
EnergyIPMICALCAdjustment=no      #approximate for more accurate task consumption

EnergyIPMIUsername=USERNAME
EnergyIPMIPassword=PASSWORD
```

These are configurations that only impact:

```
AcctGatherEnergyType=acct_gather_energy/ipmi
```

Enabling Power Collection

AcctGatherFilesystemType

One last unique collection is Filesystem type, of which only one type exists: Lustre.

If you do not have Lustre, this feature will be of no use and can be left undefined.
When enabled, it will report I/O metrics:

Reads, ReadMB, Writes, WriteMB

Enabling Power Collection

AcctGatherProfileType

An API call can probe continuously for this information and it can be stored in a place of your choice, but alternatively, you can set `AcctGatherProfileType` to have much more structured information saved—or even sent to an InfluxDB database:

```
AcctGatherProfileType=acct_gather_profile/hdf5
```

Have hdf5 files created on the filesystem (requires hdf5 in the compile!)

```
AcctGatherProfileType=acct_gather_profile/influxdb
```

Send data to an InfluxDB (time-series) database.

Choosing a `AcctGatherProfileType` like this is not required.

Enabling Power Collection

AcctGatherProfileType

When enabling AcctGatherProfileType, the following information becomes available for capture:

- Energy data (power usage)
- File system (Lustre) data, e.g., Reads, ReadMB, Writes, WriteMB
- Network (InfiniBand) data, e.g., PacketsIn, PacketsOut, InMB, OutMB
- Task (I/O, Memory, ...) data

Enabling Power Collection

`acct_gather_interconnect`

Lastly, if you have an interconnect and are interested in the ingress and egress, consider turning on this collection mechanism:

```
AcctGatherProfileType=acct_gather_interconnect/OFED
```

For those using infiniband, OFED is available.

```
AcctGatherProfileType=acct_gather_profile/sysfs
```

And Sysfs, present in all Linux systems, for ordinary interfaces like ethernet.

Choosing to include `acct_gather_interconnect` again is not required.

Power Usage and Monitoring

Thank you!

Any questions?